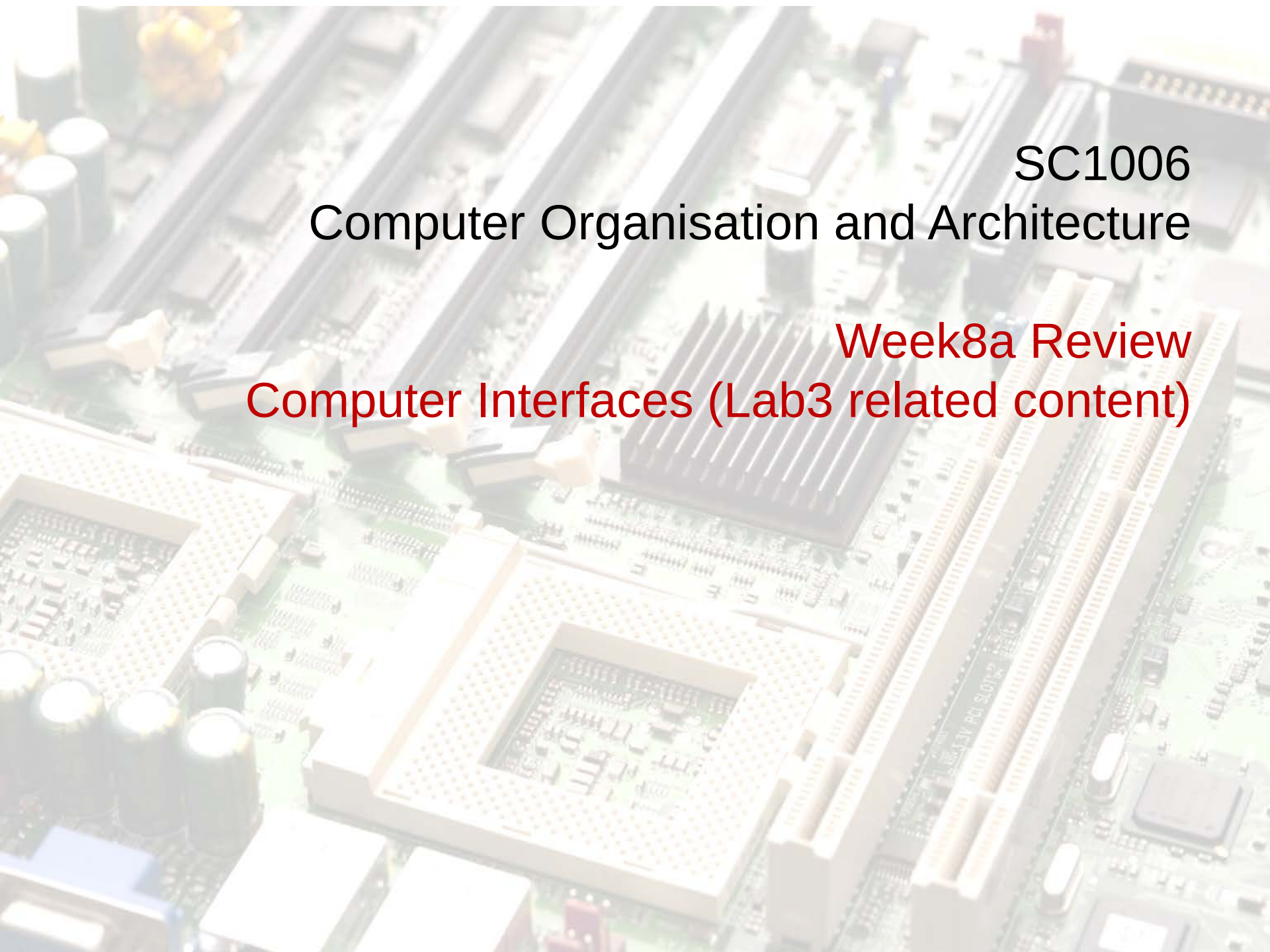


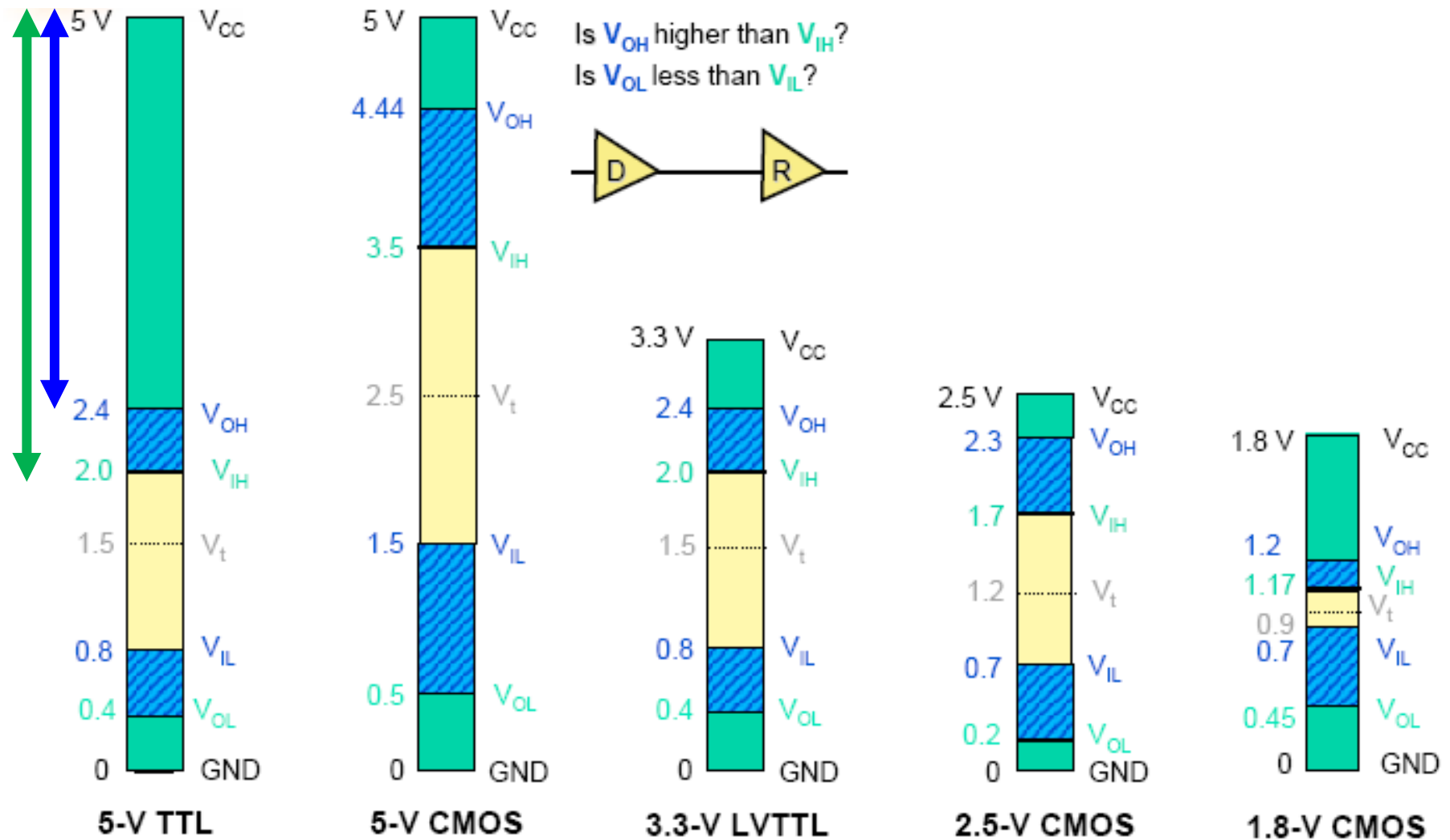


SC1006

Computer Organisation and Architecture

Week8a Review
Computer Interfaces (Lab3 related content)

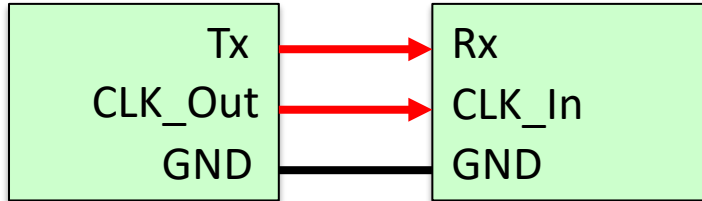
Electrical Signal Level Standards



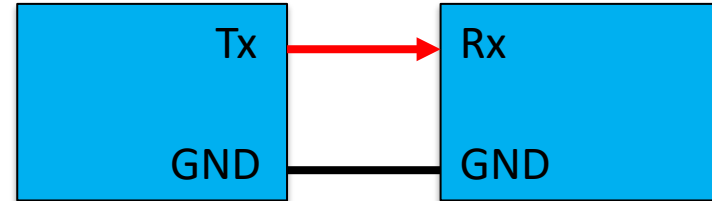
Source: http://denethor.wlu.ca/pc200/logic_charac/voltage_char.png

Synchronous vs Asynchronous

Synchronous

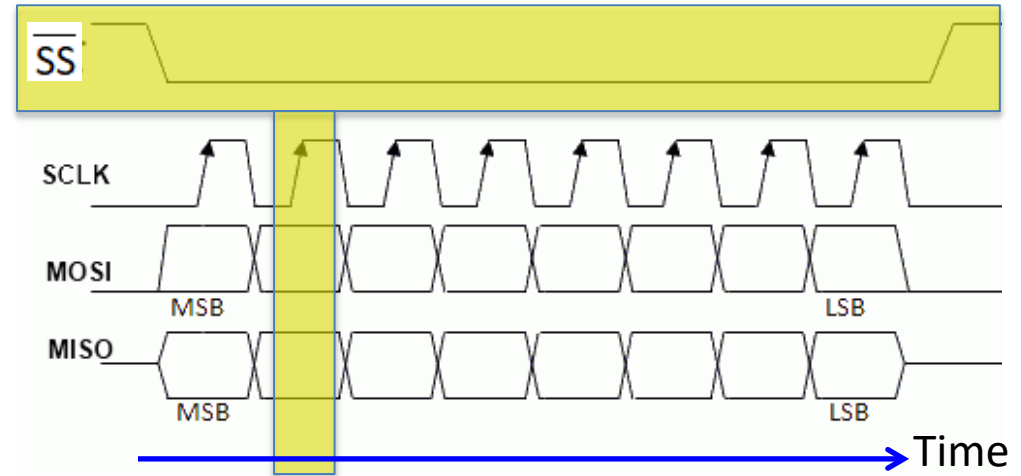
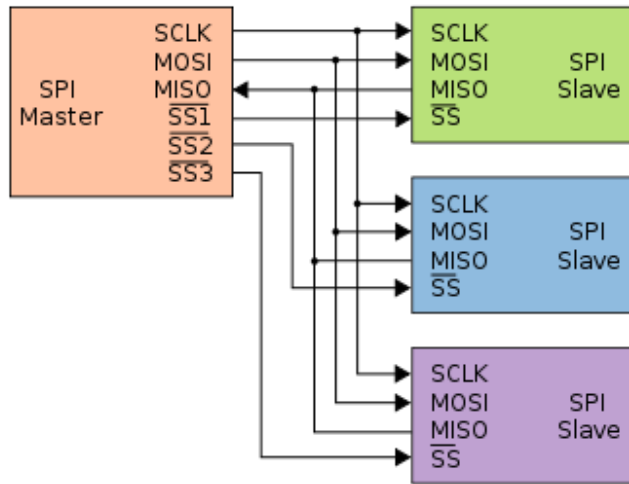


Asynchronous



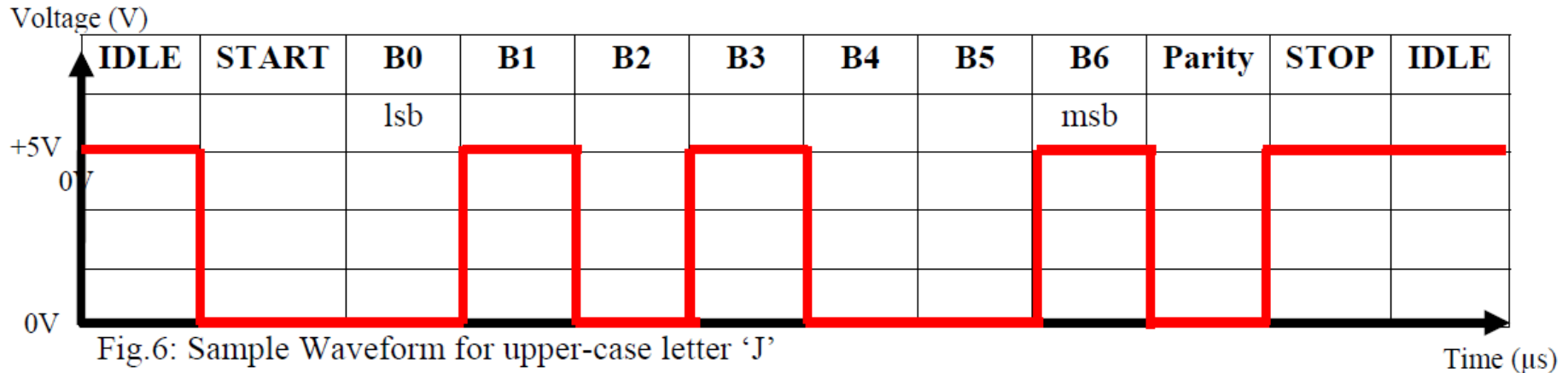
- If there is a **common clock signal** between the Transmitter and Receiver, then the communication is termed **synchronous**. Else, the communication is termed **asynchronous**.
- In **synchronous** transmission, there is a **common clock** signal to synchronize the data transfer. E.g. receiver to latch in the data at every rising edge of the clock.
- In **asynchronous** transmission, there is **no common clock** signal so devices have to agree on a pre-fixed clock frequency to use for data transfer.

Serial Peripheral Interface (SPI) Bus



- To **start** the transfer
 - Slave Select (**SS**) has to be pulled **Low**.
- **Data transfer**
 - **Data** on MOSI and MISO **latched** in on **rising/falling** clock edge (configurable)
 - MOSI: Master-Out-Slave-In (Master Output, Slave Input)
 - MISO: Master-In-Slave-Out (Slave Output, Master Input)
- Allow **multiple slaves** via use of multiple Slave Select.

UART Tx Example

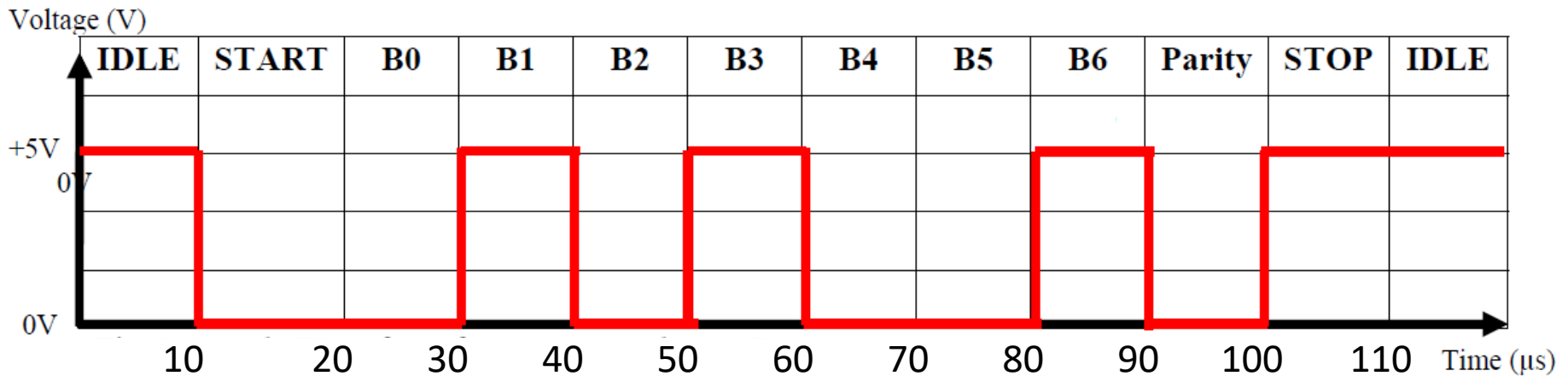


- Figure 6 above correspond to the configuration **7O1** (7 Data, 1 STOP and Odd parity).
- Capital Letter 'J' => Ascii value = 0x4A
- 0x4A => 100 1010 (binary)
- Presented in LSB first => 0101 001
- IDLE=1, START=0, STOP=1.
- Parity bit logic depends on number of 1's in Data Field and the Parity Scheme (Odd or Even) used. It can be of No parity check.

Can practise with **8E1**, **7N1**, **7N2**, ...

UART Rx Example

Tx baud rate = $1/(10 \text{ us}) = 100000 \text{ bps}$. Configuration = 701.



- RX Data = 1001010b
- RX Data = 0011000b always ?
- **Information Sampled @ 200000 bps:**
00001100110000110011